

✓ Software Testing Detailed Notes with Interview Prep

1. Test Plan

Summary from Transcript

- A **Test Plan** is a document that defines the **scope, strategy, objectives, schedule, resources, and deliverables** for testing.
- Includes **what to test, what not to test, test strategy, roles & responsibilities, entry/exit criteria, risk mitigation, and approvals.**

Additional Details

- **Purpose:** Provides a roadmap for testing activities.
- **Contents of Test Plan:**
 - Overview
 - Scope (Inclusions/Exclusions)
 - Test Strategy (Manual/Automation, Smoke, Regression)
 - Roles & Responsibilities
 - Schedule & Timelines
 - Deliverables (Test cases, reports)
 - Entry & Exit Criteria
 - Suspension & Resumption Criteria
 - Tools to be used
 - Risks & Mitigation
 - Approvals

Diagram:

[Test Plan]

└— Scope

└— Strategy

└— Schedule

└— Resources

└— Deliverables

Real-world Example:

For an **e-commerce app**, the test plan specifies:

- Scope: Functional testing of cart & payment
- Exclusions: Performance testing
- Strategy: Manual + Selenium automation
- Schedule: 2 weeks regression cycle

Interview Importance

- Common question for QA roles.

Possible Interview Questions:

- What is a Test Plan?
 - What does a Test Plan contain?
 - How do you prepare a Test Plan?
 - Give an example of a Test Plan for a banking app.
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2. Use Case vs Test Scenario vs Test Case

Summary from Transcript

- **Use Case:** Describes a requirement with **Actor → Action → Outcome**.
- **Test Scenario:** Represents **what to test**.
- **Test Case:** Represents **how to test** (step-by-step actions).

Additional Details

- **Use Case:** Created by Business Analyst; helps understand requirements.
- **Test Scenario:** Derived from use cases; high-level testing area.
- **Test Case:** Detailed steps, expected results, preconditions.

Diagram:

Use Case → Test Scenario → Test Case

Real-world Example:

- **Use Case:** User logs in successfully.

- **Test Scenario:** Validate login functionality.
- **Test Cases:**
 1. Login with valid credentials
 2. Login with invalid password
 3. Login without username

Interview Importance

- Frequently asked to check conceptual clarity.

Possible Interview Questions:

- Difference between Use Case and Test Case?
- How do you derive Test Scenarios from Use Cases?
- Write test cases for a login page.

3. Test Case Template

Summary from Transcript

- Test cases are documented in Excel or Test Management tools.
- Contains **ID, Title, Description, Preconditions, Steps, Expected Result, Actual Result, Priority, Requirement ID.**

Additional Details

Sample Template:

Test Case ID	Title	Preconditions	Steps	Expected Result	Actual Result
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Real-world Example:

Login Test Case:

- ID: TC001
- Title: Validate login with valid credentials
- Steps:
 1. Open URL
 2. Enter username/password
 3. Click Login

- Expected: Navigate to dashboard

Interview Importance

- Often tested practically.

Possible Interview Questions:

- What are the components of a Test Case?
 - Write a test case for a search functionality.
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4. Requirement Traceability Matrix (RTM)

Summary from Transcript

- RTM maps **Requirement IDs** to **Test Case IDs** to ensure coverage.

Additional Details

Purpose:

- Ensures all requirements are tested.
- Prevents missing functionality.

Diagram:

Requirement ID → Test Case IDs

Real-world Example:

Req ID	Description	Test Cases
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R001	Login	TC001, TC002
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Interview Importance

- Common question for coverage analysis.

Possible Interview Questions:

- What is RTM?
 - Why do we need RTM?
 - Show an example RTM.
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5. Test Environment Setup & Test Execution

Summary from Transcript

- Test environment replicates **customer environment**.
- Test execution involves running test cases, logging results, and reporting defects.

Additional Details

Environment Includes:

- Hardware (RAM, CPU)
- Software (OS, DB)
- Network configurations

Diagram:

[Test Environment]

└─ Hardware

└─ Software

└─ Network

Real-world Example:

Banking app tested on Windows 10, SQL Server DB, 8GB RAM.

Interview Importance

- Asked to check practical knowledge.

Possible Interview Questions:

- What is a Test Environment?
- How do you set up a Test Environment?
- What is Test Bed?

6. Defects/Bugs

Summary from Transcript

- Defect = mismatch between expected and actual result.
- Reported via tools like **JIRA, Bugzilla**.

Additional Details

Defect Lifecycle:

New → Assigned → Fixed → Retested → Closed

Diagram:

New → Open → In Progress → Fixed → Closed

Real-world Example:

Bug: Login button not working → Reported in JIRA with steps & screenshot.

Interview Importance

- Highly common question.

Possible Interview Questions:

- What is a defect?
 - Explain defect lifecycle.
 - Which tools have you used for defect tracking?
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7. Defect Report Contents

- Defect ID
 - Description
 - Steps to reproduce
 - Severity & Priority
 - Status
 - Attachments (screenshots, logs)
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8. Defect Classification (Severity & Priority)

Summary from Transcript

- **Severity:** Impact on functionality/business.
- **Priority:** How soon it should be fixed.

Additional Details

Severity Types:

- Blocker

- Critical
- Major
- Minor

Priority Levels:

- P0 (High)
- P1 (Medium)
- P2 (Low)

Examples Table:

Scenario	Severity	Priority
Login not working	Blocker	High
Spelling mistake on homepage	Minor	High
Crash in rare case	Critical	Low

Interview Importance

- Very common scenario-based question.

Possible Interview Questions:

- Difference between Severity and Priority?
- Give examples of High Severity & Low Priority defects.
- Explain combinations of Severity & Priority.

✅ Quick Revision Table for Interview

Topic	Key Points	Common Questions
Test Plan	Scope, Strategy, Deliverables	What does a Test Plan contain?
Use Case vs Test Case	Use Case = Requirement, Test Case = Steps	Difference between Use Case & Test Case
RTM	Maps Req → Test Cases	Why RTM is needed?
Defect Lifecycle	New → Closed	Explain defect lifecycle

Topic	Key Points	Common Questions
Severity vs Priority	Impact vs Urgency	Give examples
